NPRN: Wheeled Biped with Direct Readout

Mathematical description of the plant and of the network

Source: `NPRN_biped_(direct_readout)_v5.html`

The simulation couples a *wheeled inverted pendulum* whose leg length and mass change without notice (Part 1) to a *growing bank of tiny recurrent predictors* whose winner's hidden state is read out directly as a control torque (Part 2). Part 1 contains everything physical. Part 2 treats the plant as a black box that emits a 2-D observation x_t and accepts a scalar command u_t.

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1 Wheeled biped dynamics

1.1 Coordinates, constants, and time base

The robot is a wheel of radius r_w carrying a body that is modelled as a point mass m at distance l along a leg that pivots about the wheel axle. Generalised coordinates are the wheel-centre position s and the pitch θ (zero when upright, positive when leaning forward). Time is advanced with explicit Euler substeps of $\delta = 0.01\,\mathrm{s}$, five per control *tick*, so the tick period is $\Delta t = 0.05\,\mathrm{s}$ (20 Hz). All commands are held constant over the five substeps of a tick.

Symbol	Value	Symbol	Value
wheel radius r_w	0.08 m	gravity g	9.81 m/s^2
wheel mass m_w	1 kg	body inertia factor I_B	0.01 m^2
torque limit $\tau_{\max}$	2.5 N m	arena wall S_w	2.6 m
fall threshold θ_f	1.1 rad	substeps per tick	5

1.2 Equations of motion

Define the effective translating mass and the pitch inertia

$$M = m + 1.5\,m_w, \qquad J = m\,(l^2 + I_B). \tag{1}$$

The factor $1.5\,m_w$ is the wheel's translational plus rolling (solid-disk) inertia. The wheel torque τ acts on the wheel, and its reaction acts on the body. With $c = \cos\theta$, $\sigma = \sin\theta$, and a horizontal gust force F on the body, the dynamics are the 2×2 Lagrangian system

$$\underbrace{\begin{bmatrix} M & m\,l\,c \\ m\,l\,c & J \end{bmatrix}}_{\mathcal{M}(\theta)} \begin{bmatrix} \ddot{s} \\ \ddot{\theta} \end{bmatrix} = \begin{bmatrix} r_1 \\ r_2 \end{bmatrix}, \tag{2}$$

$$r_1 = \frac{\tau}{r_w} + F + m\,l\,\sigma\,\dot{\theta}^2 - 2\,m\,\dot{l}\,c\,\dot{\theta}, \tag{3}$$

$$r_2 = -\tau + F\,l\,c + m\,g\,l\,\sigma - 2\,m\,l\,\dot{l}\,\dot{\theta}. \tag{4}$$

The terms in $\dot{l}$ are Coriolis terms from leg extension: $-2ml\dot{l}\,\dot{\theta}$ is the loss of pitch rate required by conservation of angular momentum when the leg lengthens. Mass drift is slow and enters only parametrically, with no $\dot{m}$ terms.

Inverting $\mathcal{M}$ in closed form gives the accelerations that are actually integrated:

$$\ddot{s} = \frac{J r_1 - mlc\,r_2}{D}, \qquad \ddot{\theta} = \frac{M r_2 - mlc\,r_1}{D}, \qquad D = MJ - m^2 l^2 c^2. \tag{5}$$

Then $v = \dot{s}$, $\omega = \dot{\theta}$, and the wheel angle follows $\dot{\varphi}_w = v/r_w$. If $|s|$ reaches S_w the position is clamped to the wall and $v \leftarrow 0$.

1.3 Gust disturbance

The force F is an Ornstein–Uhlenbeck process, a genuine horizontal force on the body (not a state kick):

$$dF = -\frac{F}{0.2}\,dt + 3.5\,dW_t, \qquad \mathrm{Var}[F_\infty] = \tfrac{1}{2} \cdot 3.5^2 \cdot 0.2 = 1.225\ \mathrm{N}^2\ (\sigma_F \approx 1.11\ \mathrm{N}), \tag{6}$$

discretised as $F \leftarrow F - 5F\,\delta + 3.5\sqrt{\delta}\,\xi$, $\xi \sim \mathcal{N}(0,1)$. The "Push" button adds an angular-velocity impulse $\Delta\omega = \pm 1.2/(ml)$.

1.4 Actuation: network command plus safety fallback

The torque applied to the wheel is the network command u_t (Part 2) plus an optional fixed PD fallback acting on the *true* pitch error, then saturated:

$$\tau = \mathrm{sat}_{\tau_{\max}}\Big(u + \kappa\,[\,K_\theta(\theta - \theta_r) + K_\omega\,\omega\,]\Big), \qquad K_\theta = 18,\ K_\omega = 2.2, \tag{7}$$

with $\kappa = 0.4$ when the fallback is on and $\kappa = 0$ when it is off. The effective fallback gains are therefore (7.2, 0.88). A saturation flag $\texttt{sat} = \mathbf{1}[\,|u + \ldots| \geq \tau_{\max}\,]$ is exposed to the learner. Torque is positive in the direction that accelerates the wheel forward and tilts the body *backward* (note the $-\tau$ in r_2), so positive feedback on θ is stabilising.

1.5 Lean reference and arena

A drive command $v_{\mathrm{cmd}} \in [-1, 1]$ (the pad), an optional sinusoidal "rocking" term, a centring term when the pad is released, and a wall-repulsion term $w(s)$ are combined into a lean reference through a proportional law on the speed error:

$$\theta_r = \mathrm{clip}\Big(0.2\,\big[\,1.5\,v_{\mathrm{cmd}} + 0.3\,\sin(0.5\,t)\,\mathbf{1}_{\mathrm{rock}} - 0.5\,s\,\mathbf{1}[v_{\mathrm{cmd}} = 0] + w(s) - v\,\big],\ -0.3,\ 0.3\Big), \tag{8}$$

$$w(s) = -1.5\ \mathrm{sign}(s)\,\big(|s| - 2.2\big)\,\mathbf{1}[\,|s| > 2.2\,]. \tag{9}$$

1.6 Hidden morphology schedule

Four morphologies $(l^\star, m^\star)$ are visited cyclically in the order $0, 1, 2, 0, 1, 3, 2, 0, 3, 1$, each for 1200 ticks (60 s). When a new morphology starts, leg length and mass relax exponentially toward its targets at a rate of 0.004 per tick:

$$\dot{l} = 0.08\,(l^\star - l)\ \ \mathrm{s}^{-1}, \qquad m \leftarrow m + 0.004\,(m^\star - m)\ \text{per tick}, \tag{10}$$

giving a time constant of 250 ticks (12.5 s). In manual mode the leg height target is the pad's vertical coordinate $h^\star \in [0.25, 0.60]$ and the relaxation gain becomes 0.02.

index	$l^\star$ [m]	$m^\star$ [kg]	description
0	0.40	4.0	nominal
1	0.60	2.5	tall, light
2	0.25	7.0	short, heavy
3	0.60	7.0	tall, heavy

1.7 Failure and recovery

If $|\theta| > \theta_f = 1.1$ the fall counter increments, the pitch is set to $\mathrm{sign}(\theta)\,1.25$, and $\omega \leftarrow 0$. For the next 20 ticks (1 s) the robot lies down: the wheel free-rolls ($v \leftarrow 0.99\,v$ per substep), no command is applied, and the network is stopped. After that $\theta = \omega = v = 0$ (self-righted).

1.8 Linearised analysis

Linearising (2) about $\theta = \omega = \dot{l} = F = 0$ and eliminating $\ddot{s}$ gives the pitch subsystem

$$\ddot{\theta} = p^2\,\theta - b\,\tau + (\mathrm{gust}), \qquad p^2 = \frac{M\,g\,l}{(M - m)\,l^2 + M I_B}, \qquad b = \frac{M + m\,l/r_w}{m\,[(M - m)\,l^2 + M I_B]}. \tag{11}$$

The open-loop pole $+p$ is the instability rate, and b is the torque-to-pitch-acceleration gain. Both depend on (l, m), which is exactly what the controller cannot observe. Two derived quantities are informative:

- **Recoverable lean angle.** With the torque limit, the largest lean from which a purely pitch-restoring torque can win against gravity is roughly $\theta_{\text{rec}} \approx b\,\tau_{\max}/p^2$.
- **Fallback loop.** With $u = 0$ and $\kappa = 0.4$, $\ddot{\theta} + 0.88\,b\,\dot{\theta} + (7.2\,b - p^2)\,\theta = 0$. The natural frequency is $\omega_n = \sqrt{7.2\,b - p^2}$ and the damping ratio is $\zeta = 0.88\,b/(2\omega_n)$.

morph	l [m]	m [kg]	p [s^{-1}]	b	θ_{rec} [rad]	ω_n	ζ
0	0.40	4.0	8.55	21.6	0.74	9.08	1.05
1	0.60	2.5	6.37	15.7	0.97	8.51	0.81
2	0.25	7.0	10.80	24.3	0.52	7.63	1.40
3	0.60	7.0	8.95	13.9	0.44	4.51	1.36

The fallback alone makes the pitch loop stable and well damped in all four morphologies (each $7.2\,b > p^2$). The heavy, tall robot (morphology 3) has the smallest recoverable angle and the lowest bandwidth, so it is the hardest regime. This analysis ignores the 20 Hz zero-order hold, the lean reference, and the cart translation mode.

1.9 Sensor interface

The only signal that leaves the plant for the network is the noisy, rescaled tracking-error pair

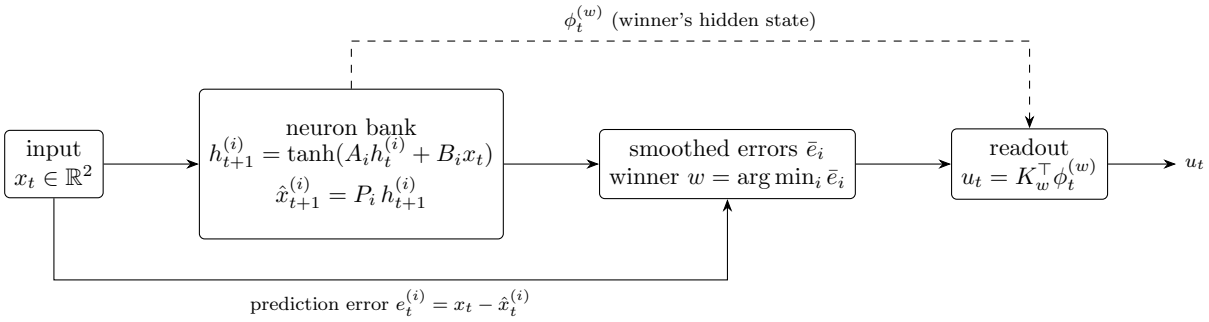
$$x_t = \begin{bmatrix} 5\,(\theta_t - \theta_{r,t}) \\ 0.5\,\omega_t \end{bmatrix} + 0.01\,\xi_t, \qquad \xi_t \sim \mathcal{N}(0, I_2), \tag{12}$$

sampled once per tick after the plant step. The network never observes l, m, s, v, or F, and it returns only the scalar u_t in (7).

2 Neural network architecture

2.1 Interface and data flow

The network is a variable-size set of $N_t \le N_{\max} = 40$ neurons $\{\mathcal{N}_i\}$. At every tick it receives $x_t \in \mathbb{R}^2$ and emits $u_t \in \mathbb{R}$. Every neuron sees the same input. A hard winner-take-all selects the neuron that currently predicts best, and only that neuron's control readout drives the plant.



2.2 A single neuron

Neuron i stores fixed reservoir weights (A_i, B_i), learned readouts (P_i, K_i), a hidden state $h_t^{(i)}$, and bookkeeping scalars (smoothed error $\bar{e}_i$, age a_i, utilisation ν_i, probation counter π_i, win counts per morphology).

Hidden dynamics (fixed weights). The hidden state is $h_t^{(i)} \in \mathbb{R}^2$:

$$h_{t+1}^{(i)} = \tanh\big(A_i\, h_t^{(i)} + B_i\, x_t\big), \quad A_i = 0.25\, I + 0.03\, \mathcal{G}, \quad B_i = 0.6\, I + 0.03\, \mathcal{G}, \tag{13}$$

where $\mathcal{G}$ denotes a matrix of i.i.d. $\mathcal{N}(0,1)$ entries drawn independently for each neuron. A new neuron starts from $h_0^{(i)} = \tanh(0.6\, x)$. Since $\rho(A_i) \approx 0.25$, the cell is strongly contractive with fading memory of about one tick, so it behaves nearly like the static feature map $h \approx \tanh(0.6\, x)$ plus a small recurrent correction. A_i and B_i are *never trained*.

Predictive readout (learned). With $P_i \in \mathbb{R}^{2\times 2}$, $P_i(0) = 0$:

$$\hat{x}_t^{(i)} = P_i\, h_t^{(i)}, \qquad e_t^{(i)} = x_t - \hat{x}_t^{(i)}, \qquad \varepsilon_t^{(i)} = \|e_t^{(i)}\|_2. \tag{14}$$

Because h_t is a function of x_{t-1} and earlier inputs, $\hat{x}_t$ is a genuine one-step-ahead prediction.

Control readout (learned). With $K_i \in \mathbb{R}^3$, $K_i(0) = 0$, and a feature vector that appends a constant bias input:

$$\phi_t^{(i)} = \begin{bmatrix} h_{t,1}^{(i)} \\ h_{t,2}^{(i)} \\ 0.2 \end{bmatrix}, \qquad u_t^{(i)} = K_i^\top \phi_t^{(i)}. \tag{15}$$

Each neuron therefore carries 15 numbers: 8 fixed (A_i, B_i) and 7 learned (P_i, K_i).

2.3 Competition and the direct readout

Each neuron's error is smoothed with a fast exponential filter, and the winner is the smallest:

$$\bar{e}_i \leftarrow 0.8\, \bar{e}_i + 0.2\, \varepsilon_t^{(i)}, \qquad w_t = \arg\min_i\ \bar{e}_i, \qquad u_t = K_w^\top \phi_t^{(w)}. \tag{16}$$

The command u_t is computed from the same hidden state that does the predicting, so there is no separate policy network. The control is a dynamic output feedback $u = K_w^\top[\tanh(A_w h + B_w x),\ 0.2]$ built on the predictor's state. Since $\bar{e}_i$ depends on the closed-loop observation statistics, the winner index w_t acts as an online regime estimate, and K_w plays the role of a regime-specific gain set. This is multiple-model adaptive control with a learned model bank.

2.4 Learning the predictors: normalised LMS

For neuron i with current $h = h_t^{(i)}$ and error $e = e_t^{(i)}$:

$$P_i \leftarrow P_i + \frac{\eta_i}{\epsilon + \|h\|^2}\, e\, h^\top, \qquad \epsilon = 0.02, \qquad \eta_i = 0.4 \cdot \mathrm{clip}\Big(\frac{\bar{e}_i}{\vartheta},\ 0.2,\ 2\Big), \tag{17}$$

where ϑ is the spawn/learn threshold (default 0.1, slider range 0.02–0.3). This is stochastic gradient descent on $\frac{1}{2}\|e\|^2$ with NLMS normalisation. If $P^\star$ is the best linear map and $\tilde{P} = P - P^\star$, the noise-free update is

$$\tilde{P}^+ = \tilde{P}\Big(I - \frac{\eta_i\, hh^\top}{\epsilon + \|h\|^2}\Big), \tag{18}$$

whose factor has eigenvalue 1 on $h^\perp$ and $1 - \eta_i\|h\|^2/(\epsilon + \|h\|^2) \in (1 - \eta_i, 1)$ along h. The update is therefore a contraction whenever $0 < \eta_i < 2$, and here $\eta_i \le 0.8$.

Gating. Only a subset of neurons is allowed to learn at each tick:

$$\mathcal{L}_t = \{\, i :\ \pi_i > 0 \,\} \ \cup\ \{\, i = w_t :\ \bar{e}_i < 1.5\,\vartheta \,\}. \tag{19}$$

A neuron is in *probation* for its first $\pi_i = 400$ ticks (learning unconditionally). After that it learns only if it is the current winner *and* fits well, so a neuron that does not match the present regime keeps its parameters instead of overwriting them. This is the stability–plasticity mechanism: novelty is handled by spawning (Section 2.6), not by overwriting.

2.5 Learning the control readout: sign-prior NLMS

Only the winner's K_w is updated, using a tracking signal that is a fixed linear function of the (noise-free) observation:

$$y_t = (\theta_t - \theta_{r,t}) + 0.15\,\omega_t = 0.2\,x_{t,1} + 0.3\,x_{t,2} \ \ \text{(up to sensor noise)}, \tag{20}$$

$$K_w \leftarrow \mathrm{clip}\Big(K_w + \mu\,\frac{y_t}{\epsilon + \|\phi\|^2}\,\phi,\ -15,\ 15\Big), \qquad \mu = 0.3 \ \ \text{(slider 0.02–1.5)}. \tag{21}$$

The rule is gradient descent on $\frac{1}{2}y^2$ in which the unknown plant sensitivity is replaced by a *sign prior*, $\partial y/\partial u < 0$, so that

$$\Delta K = -\mu\, y\, \frac{\partial y}{\partial K} \ \approx\ +\mu\, y\, \frac{\phi}{\epsilon + \|\phi\|^2}. \tag{22}$$

No plant model is needed: the gain simply pushes u in the direction of y. The update is evaluated after the plant step and is skipped whenever the plant is saturated or fallen, the network is off, or the winner is neither in probation nor fitting ($\bar{e}_w \ge 2\,\vartheta$). Skipping on saturation prevents wind-up when the demanded torque exceeds the actuator limit. With $\|K\|_\infty \le 15$, $|u|$ can reach about 33, well above any actuator limit.

2.6 Structural plasticity

Spawn. Let nov_t count consecutive ticks with $\bar{e}_w > \vartheta$ (reset to 0 otherwise). A neuron is created when

$$\mathrm{nov}_t > 15, \qquad t - t_{\mathrm{spawn}} > 400, \qquad N_t < 40. \tag{23}$$

The new neuron has fresh random (A, B), $P = 0$, $h_0 = \tanh(0.6\, x_t)$, and a *warm-started* controller $K \leftarrow K_w$, so control quality does not reset to zero. A bootstrap spawn creates the first neuron when the bank is empty. Spawning is thus driven by sustained prediction failure of the *best* existing model.

Prune. Define utilisation ν_i as an exponential moving average of the winner indicator with time constant 8000 ticks, initialised at 0.05:

$$\nu_i \leftarrow \Big(1 - \tfrac{1}{8000}\Big)\nu_i + \tfrac{1}{8000}\,\mathbf{1}[\,i = w_t\,]. \tag{24}$$

Every 200 ticks, neurons with age $a_i > 3000$ and $\nu_i < 0.0015$ are deleted (provided $N_t > 1$).

Merge. Every 200 ticks, every pair (i, j) with $a_i, a_j \geq 800$ and

$$\|P_i - P_j\|_F < 0.06 \tag{25}$$

is collapsed. Neuron i survives with

$$P_i \leftarrow \tfrac{1}{2}(P_i + P_j), \quad K_i \leftarrow \tfrac{1}{2}(K_i + K_j), \quad \mathrm{wins}_i \leftarrow \mathrm{wins}_i + \mathrm{wins}_j, \tag{26}$$

and neuron j is deleted. Averaging P is meaningful only because all reservoirs are near-identical ($A \approx 0.25I$, $B \approx 0.6I$). The surviving neuron keeps its own (A_i, B_i). The "Freeze structure" control disables spawn, prune, and merge while learning continues.

2.7 Reuse bookkeeping

For each visit k to a morphology r_k, the dominant neuron is the one with most wins during the visit:

$$d_k = \arg\max_i \ \#\{t \in \text{visit } k : \ w_t = i\}. \tag{27}$$

The visit is flagged REUSED if $d_k = d_{k'}$ for some earlier visit k' of the same morphology, and as shared if $d_k = d_{k''}$ for a visit of a *different* morphology. Adaptation error is the mean of ε over the first 150 ticks of the visit, and steady-state error is the mean over the last 300.

2.8 Per-tick algorithm

Per tick t, in order:

1. **Command.** $u_t = K_w^\top \phi_t^{(w)}$, with $u_t = 0$ if the network is off.
2. **Plant step** with u_t (Part 1), then observe x_t via (12).
3. **Control update.** Apply (21) if its gates pass.
4. If the network is off or the robot is fallen, **stop here**: no prediction, learning, or structure change.
5. **Errors.** $e_t^{(i)} = x_t - P_i h_t^{(i)}$ for all i; update $\bar{e}_i$; set $b = \arg\min \bar{e}_i$.

6. **Predictor learning.** Apply (17) on $\mathcal{L}_t$; decrement π_i; increment a_i; update ν_i using b.
7. **Record** the win of neuron b (for reuse bookkeeping).
8. **State update.** $h_{t+1}^{(i)} = \tanh(A_i h_t^{(i)} + B_i x_t)$ for all i.
9. **Structure** (unless frozen): spawn check every tick; prune and merge every 200 ticks.
10. **Winner.** Recompute $w_{t+1} = \arg\min_i \bar{e}_i$ for the next command.

Note that u_t uses h_t, which was built from x_{t-1} and earlier, so the controller has an inherent one-tick delay relative to the latest observation.

2.9 Hyperparameters

Quantity	Symbol	Value
spawn / learn threshold	ϑ	0.1 (adjustable)
control learning rate	μ	0.3 (adjustable)
NLMS regulariser	ϵ	0.02
probation length	—	400 ticks
novelty run length	—	> 15 ticks
utilisation time constant	—	8000 ticks
prune: age, utilisation	—	> 3000, < 0.0015
merge distance	—	$\|P_i - P_j\|_F < 0.06$
maximum neurons	$N_{\max}$	40
readout clip	—	$\|K| \leq 15$

2.10 Remarks on the architecture

- **Where specialisation lives.** Since the reservoirs are almost identical, neurons differ mainly through P_i and K_i, not through the nonlinear hidden dynamics. The effective predictor is close to a first-order map $x_t \approx M x_{t-1}$ in tanh-warped coordinates.
- **Circularity.** The predicted signal x is itself a closed-loop quantity that depends on K_w. Learning K changes what P must predict, so a neuron identifies "plant plus its own controller". The gating in (17) and (21) limits the damage this coupling could otherwise cause.
- **Capacity growth.** The bank grows only on sustained prediction failure of the best model, and shrinks through pruning of unused neurons and merging of duplicates, so N_t tracks the number of distinct dynamical regimes encountered.